# When does generative AI create business value?

Independent reanalysis of a randomized professional-writing experiment, with a separate commercial adoption model.

**Recommendation:** investigate a bounded writing-workflow pilot. The historical experiment shows faster completion and better rated writing in retained respondents. A new company's investment case remains sensitive to verification effort, task mix and cost.

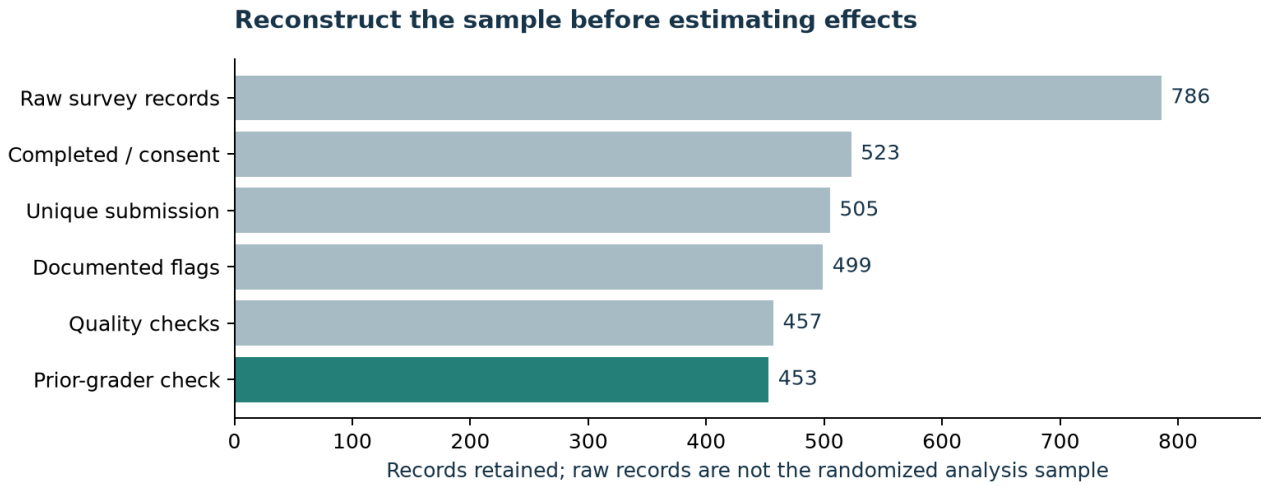
| Evidence | This project's estimate | Decision implication |
|---|---|---|
| Time, available cases | 10.34 minutes saved; 95% interval 8.10 to 12.58; N=346. | Test recurring, self-contained writing tasks first. |
| Rated quality | +0.70 points on a 1-7 scale; interval 0.47 to 0.93; N=352. | Writing scores do not establish factual accuracy. |
| Commercial base scenario | $698/month net capacity value under assumptions. | Do not present this as measured company savings. |

## What this project contributes

A reconstructed sample audit, independent Python estimators, counterbalancing checks, sensitivity analyses, exploratory baseline-score differences, SQL summaries and a transparent translation from experimental evidence to a company budget boundary.

Source experiment: Shakked Noy and Whitney Zhang, Science (2023). The BI extension preserves the original primary estimates. This project did not recruit participants or conduct the original experiment.

# Understand the experiment and the retained sample




Participants completed two occupation-specific writing tasks. Between tasks, the treatment group was offered ChatGPT access and onboarding; the control group completed a signup comparison. Task order was randomized. The experiment used the technology available in early 2023, not current AI models.

The cleaning pipeline reconstructs 453 retained participants. Primary productivity analyses use 359 participants in the linear/convex incentive arms; the fixed-time arm is excluded from the time analysis because time was constrained by design.

| Primary outcome | Available baseline/post pairs | Why the denominator differs |
|---|---|---|
| Self-reported time | 346 | Some respondents lack one or more workflow-time components. |
| Mean paired rating | 352 | Some retained submissions lack eligible paired ratings. |

Raw records include incomplete or excluded entries and must not be called the randomized analysis sample. Every filtering stage is saved in reports/data-audit.json.

# Estimate assignment effects without selecting AI users

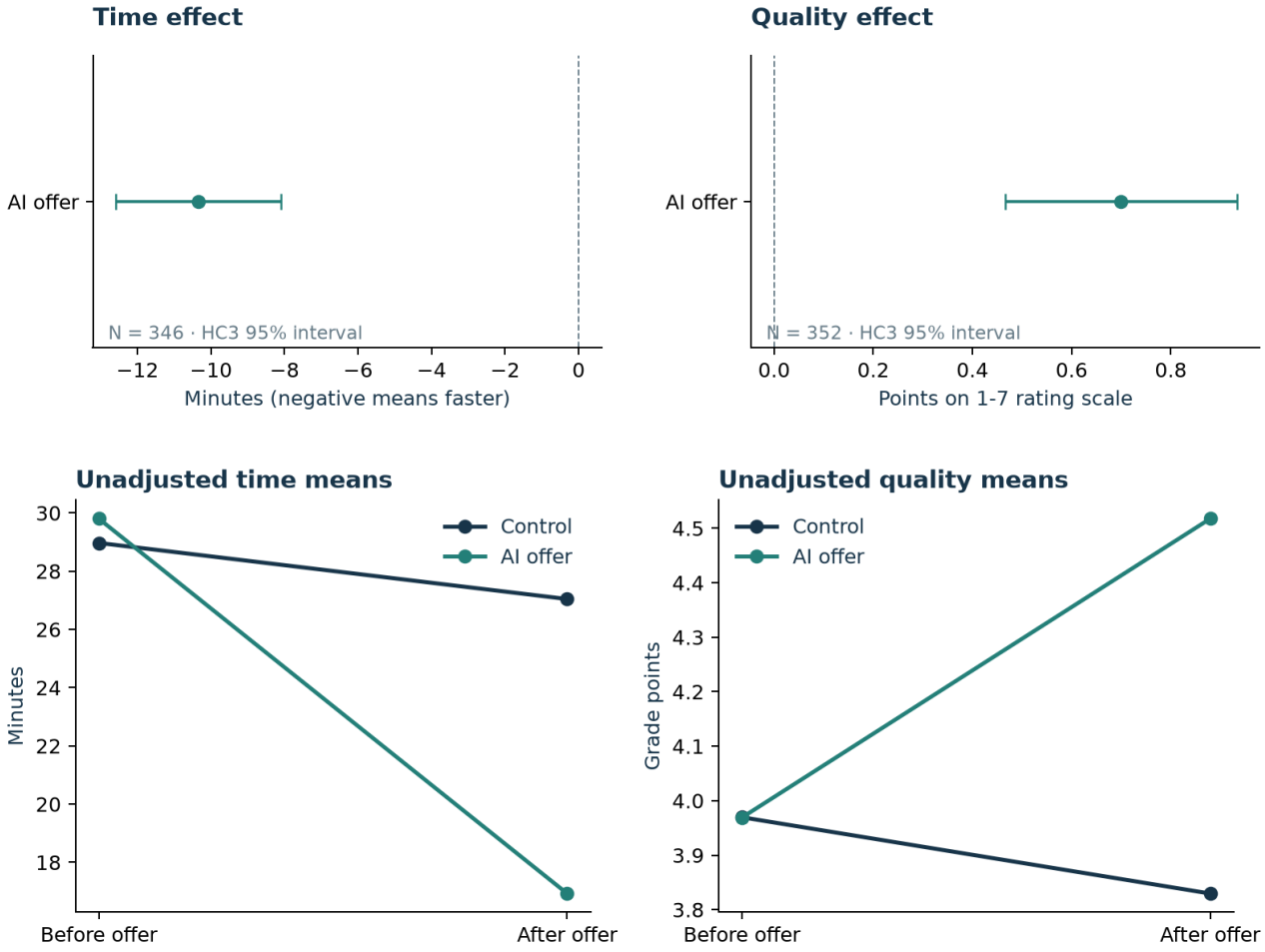
The estimand is the effect of assignment to the AI offer among retained available cases. Selecting only people who chose to use AI would condition on a post-randomization behavior and break the comparison.

| Choice | Implementation | Reason |
| --- | --- | --- |
| Counterbalancing | Map task A/B to baseline/post using the randomized task order. | Task A is not always the pre-treatment task. |
| Time outcome | Sum reported brainstorming, drafting, editing and other minutes. | Preserve the documented workflow components. |
| Quality outcome | Average valid paired grader ratings per participant. | Each participant contributes one baseline/post pair. |
| Primary adjustment | Post outcome on assignment, baseline, occupation x task order, and incentive arm. | Improve precision while respecting the study design. |
| Uncertainty | HC3 robust standard errors, t intervals; Holm adjustment for two primary tests. | Report uncertainty and limit two-outcome multiplicity. |

The study's grading model included grader-level details. This project's participant-average estimator is deliberately simpler; a paired-grader fixed-effect sensitivity model is also estimated. No exact reproduction of every published coefficient is claimed.

The full analysis plan was written before this project's models were fitted, after the original public findings were already known. It is not a new preregistration. Differential attrition limits a full-sample intention-to-treat interpretation.

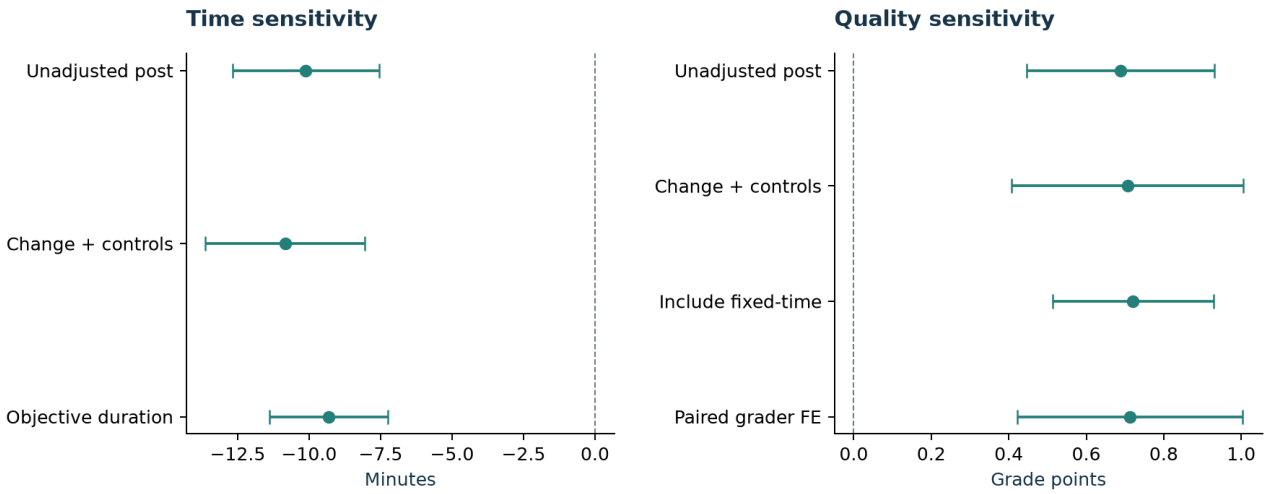
# The historical effects are sizable in both outcomes




Adjusted assignment contrast: -10.34 minutes and +0.70 quality points. Both co-primary tests remain below 0.001 after Holm adjustment. The lower chart shows descriptive available-case means, not adjusted estimates.

A grade is a rubric-based assessment of writing output, not an audited measure of factual correctness, customer impact or firm performance. Saving minutes on a study task does not establish a proportional increase in total employee productivity.

# Check whether the result survives reasonable alternatives

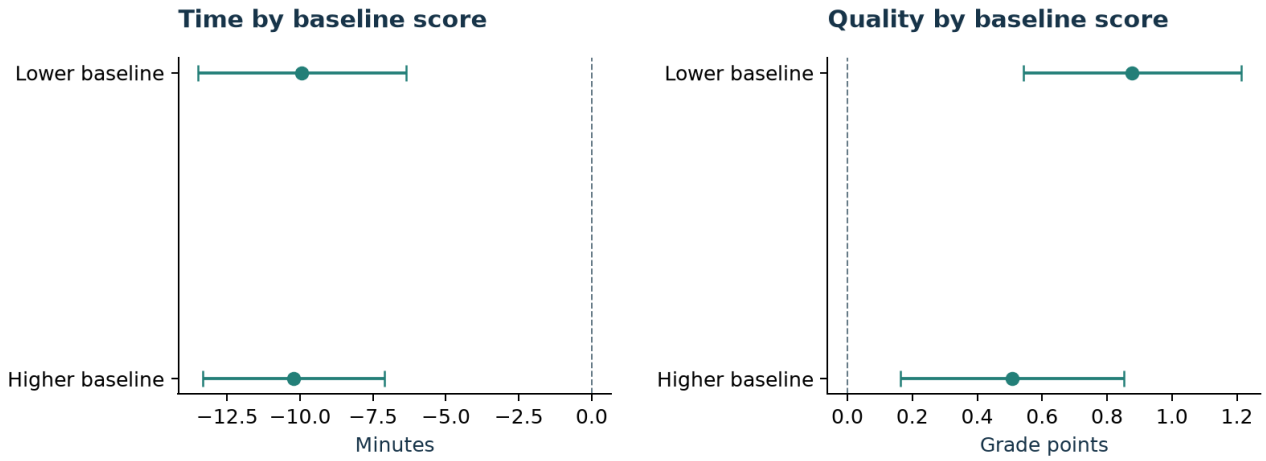



The time effect remains negative with unadjusted outcomes, change-score controls and an objective page-duration measure. The quality effect remains positive with changes, inclusion of the fixed-time arm and a paired-grader fixed-effect model with participant-clustered inference.

| Assignment | Reached first-task assessment | Completed survey | Completion rate |
|---|---|---|---|
| Control | 234 | 220 | 94.0% |
| AI offer | 218 | 195 | 89.4% |

These completion rates concern consented linear/convex participants who reached the first-task assessment. They are not the final analytic-sample counts. Differential completion and later quality exclusions remain limitations; baseline adjustment cannot guarantee their elimination.

# Quality gains vary with baseline writing performance




The quality interaction is -0.249 points per baseline writing-score point (95% interval -0.434 to -0.063). This is consistent with a larger quality improvement among lower baseline scorers in this task setting.

The time interaction is -0.417 minutes per baseline score point (interval -2.131 to 1.297); it remains imprecise. Similar time benefits in the two plotted groups should not be described as a proven subgroup difference.

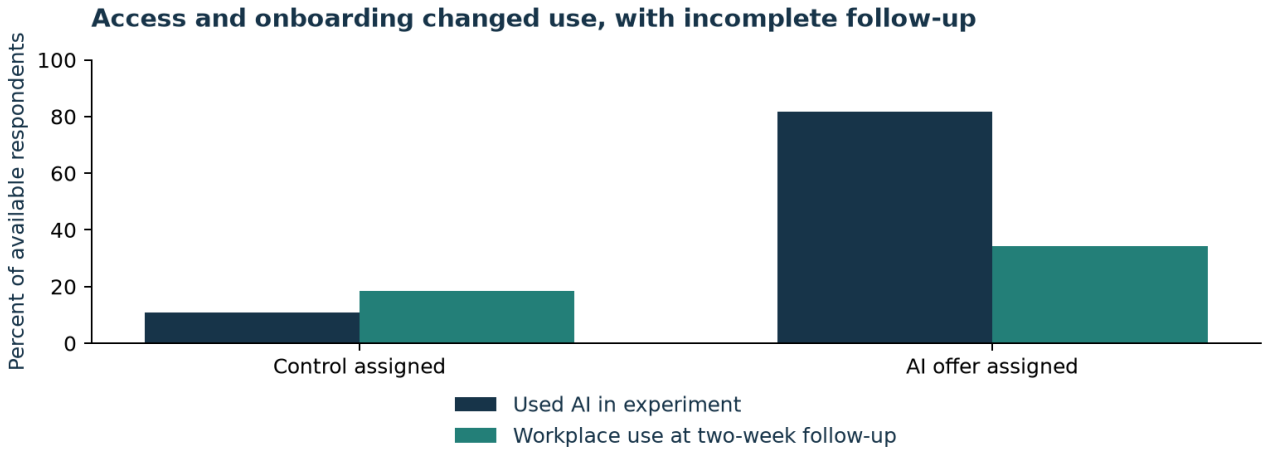
## Business interpretation

Provide onboarding broadly and investigate where writing assistance improves an existing workflow. Use task needs and source-verification requirements to scope a pilot. A single baseline writing exercise is not a general employee-ability measure and should not become a hiring or workforce-allocation score.

Group cutoffs use the pooled pre-treatment median. These extensions are exploratory, with small subgroups and no claim of an independent confirmatory subgroup experiment.

The two exploratory interaction tests have Holm-adjusted p-values 0.632 (time) and 0.018 (quality). A single noisy baseline and the 7-point ceiling remain limitations; pointwise baseline-score contrasts are shown on page 13.

# Adoption persists, but external validity remains limited




Among 316 available two-week follow-up respondents in the primary arms, the assignment contrast in reported workplace use is 15.7 percentage points (95% interval 5.9 to 25.4). This is a response-based, self-reported result; it is not a measured long-run productivity effect.

The source experiment used short, self-contained writing tasks across selected occupations. Real business work can require proprietary context, exact facts, collaboration and longer review cycles. The new company's time effect, quality and maintenance burden remain unknown.

Treat this as historical evidence motivating a new pilot. Do not present the 2023 effect as an estimate of current AI products or assume that workplace adoption makes the company economically better off.

# Translate minutes into a transparent budget boundary

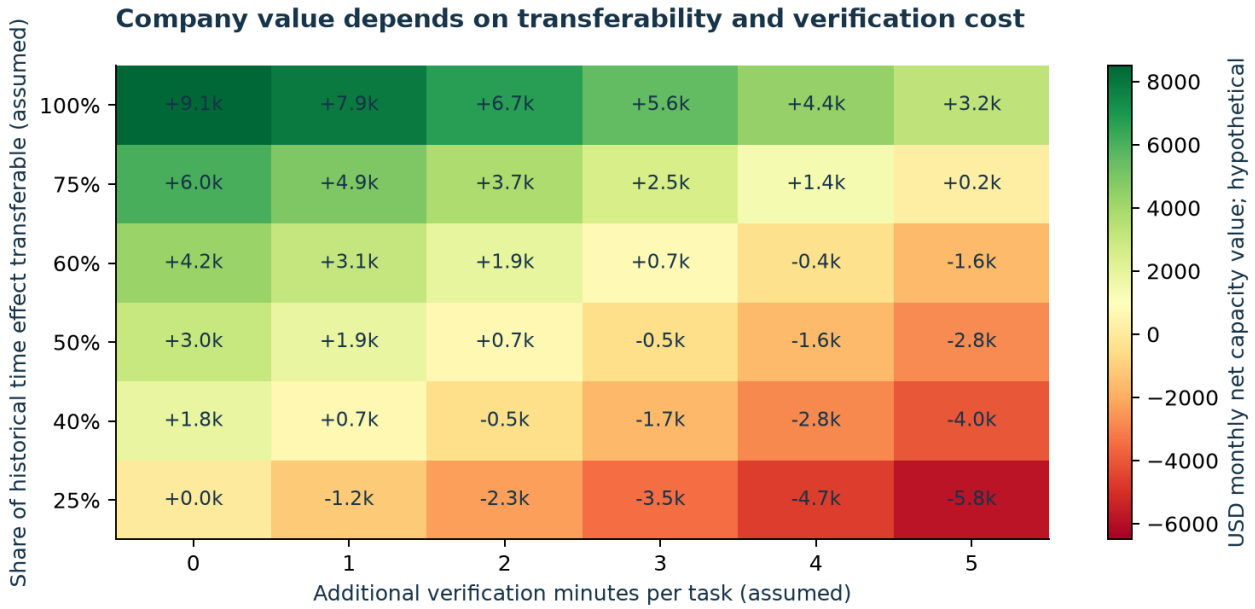
| Input | Base assumption | Status |
|---|---|---|
| Employees / rollout coverage | 200 / 50% (100 offered seats) | Company scenario |
| Eligible writing tasks | 20 per offered employee per month | Company scenario |
| Transferable historical effect | 50% of the study assignment estimate | External-validity assumption |
| Additional verification | 2 minutes per eligible task | Company scenario |
| Baseline / revision / rework | 30 minutes / 0 / 0 in this reference case | Illustrative reference inputs |
| Employee time value | $35 per hour | Capacity valuation assumption |
| Assumed license / fixed monthly cost | $25 per offered seat / $500 | Illustrative costs, not market quotes |
| One-time setup | $3,000 | Company scenario |

Gross saved minutes = min(historical assignment effect x transferability, baseline task duration). Net saved minutes subtract extra review, revision and expected rework (probability x rework duration). Capacity hours = offered seats x eligible tasks x net minutes / 60. Net capacity value subtracts recurring seat and support costs.

Base output: 105.7 hours/month of capacity, $3,698 gross time value, $3,000 recurring cost and $698 net capacity value. Simple capacity-value payback is 4.3 months under these assumptions.

The assignment effect already includes the source experiment's take-up pattern. Multiplying by that take-up rate again would double-discount the same mechanism. Offered seats incur cost even if some people do not use them.

# A strong task effect can still produce a fragile investment case




The assumed operating break-even seat cost is $31.98/month, before recovering setup cost. It is a budget ceiling under a scenario, not an observed willingness to pay or recommended market price.

Holding all commercial assumptions fixed and propagating only the statistical time-effect interval gives monthly net capacity value from $-610 to $2,007. This range crosses zero. It is not a complete confidence interval for company profit because transferability and cost assumptions have further unmeasured uncertainty.

At the base costs and task volume, additional verification above about 2.60 minutes per task removes the positive operating case. This is a concrete quantity for a pilot to measure.

Conservative, base and upside results are saved separately in reports/results.json. Released employee capacity becomes a cash saving only if a specific budget expense actually falls; that has not been measured.

# Use the evidence to design the next decision

Scope a small trial around recurring, generic writing work where factual review can be observed. Measure time to accepted output including checking and rework, blinded quality, real eligible task volume, offered-seat usage and operating cost. Stop or redesign if review burden consumes the expected gain.

| Decision | Evidence needed in the new company |
|---|---|
| Expand | Useful verified time improvement, acceptable quality and positive downside economics. |
| Revise | Benefits concentrated in a narrow task category or avoidable verification friction. |
| Stop | Quality problems, unsuitable documents, low eligible volume or costs above the budget ceiling. |

## Reproduction and original contribution

The repository contains a hash-verified downloader, a sample-flow audit, Python statistical models, SQL summaries, an executed analysis notebook, eight reproducible figures and this report. Raw and participant-level data are downloaded separately and are not republished. The source package has no declared license in its OSF metadata.

## Source study and data

Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. Science 381, 187-192. doi.org/10.1126/science.adh2586.

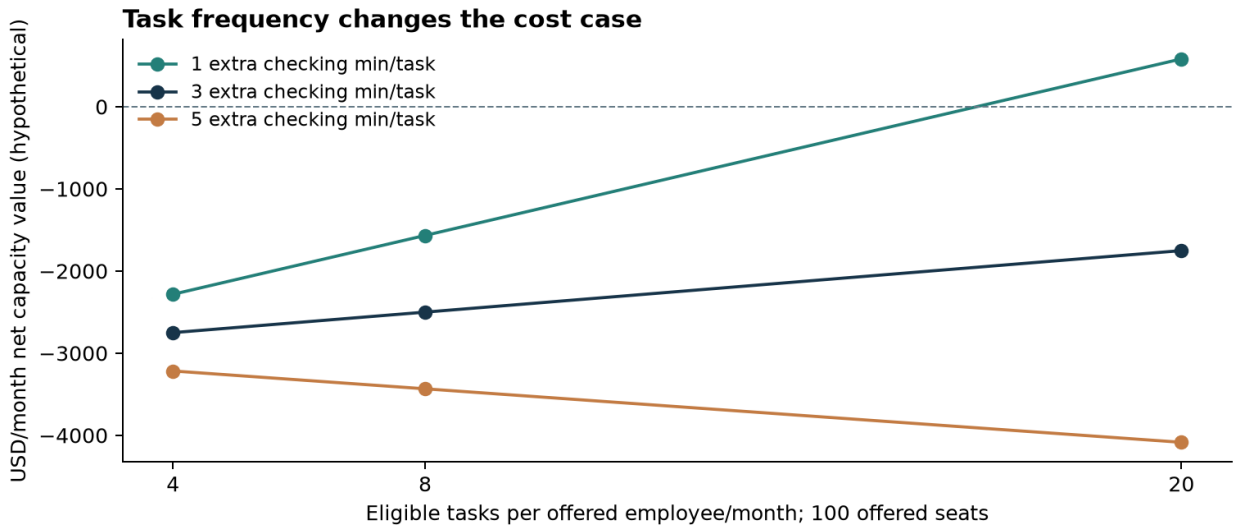
Author-hosted replication package: osf.io/xd7qw/; DOI 10.17605/OSF.IO/XD7QW. The source ReadMe documents anonymization and variable definitions.

This project: independent reanalysis, estimator comparisons and commercial scenarios. It does not claim original data collection, a current-model benchmark or realized business benefits.

# Would a weekly status email cover the tool cost?

This hypothetical extension varies task frequency and extra checking time for internal project-status writing. Other configured costs and the assumed 50% transfer of the historical effect remain fixed. No new email experiment or company data were collected.

Baseline task duration is assumed to be 10 minutes; transferred savings are capped at this duration. Each task also adds 0.5 revision minutes and expected rework of 20% x 3 minutes, beyond the selected checking time.




| Tasks/person/month | Extra checking min/task | Capacity hours/month | Net capacity value/month |
|---|---|---|---|
| 4 | 1 | 20.5 | $-2,284 |
| 20 | 1 | 102.3 | $581 |
| 20 | 3 | 35.7 | $-1,752 |

Fewer eligible tasks leave fewer opportunities to recover recurring costs. Extra checking reduces the time gained from each task. Positive time value can therefore coexist with negative net capacity value. The full nine-case SQL/Python calculation is saved with an editable configuration and narrative in the repository.

All seat/support costs are charged to this workflow; other uses of a shared subscription are not valued. Four weeks is a simplifying monthly convention. The historical saving was not measured on emails: actual task time could be too short to support the assumed saving. Measure full time to accepted output and task frequency before drawing a purchase conclusion.

# Model the data at the correct grain

The local SQLite star model contains five dimensions and two fact tables. Source IDs and participant-level warehouse records are not published. SQL performs dimension loading, wide-to-long outcome transformation, mart pivoting and KPI aggregation.

| Table | Grain | Rows |
|---|---|---|
| dim_participant | One participant | 453 |
| dim_assignment | One unique dimension value | 2 |
| dim_occupation | One unique dimension value | 6 |
| dim_incentive | One unique dimension value | 3 |
| dim_period | One unique dimension value | 2 |
| fact_outcome | Participant x baseline/post | 906 |
| fact_usage | One participant | 453 |

The outcome fact has a composite participant/period primary key. Dimension natural keys are unique and foreign keys are checked. A participant mart pivots baseline/post measurements to one row before joining usage, preventing an unqualified fact-to-fact join from multiplying measures.

| Integrity check | Violations |
|---|---|
| dimension_join_row_expansion | 0 |
| participant_mart_grain | 0 |
| missing_or_extra_measurement_period | 0 |
| assignment_changes_between_periods | 0 |
| usage_join_count | 0 |
| outcome_grain_duplicates | 0 |

A validated snapshot replaces the prior database atomically. Repeated runs rebuild rather than append. Tests cover duplicate natural keys, fact grain, foreign-key failures and preservation of the prior snapshot on invalid reload. Full schema and architecture diagrams are in the repository.

# Verify computation and qualify subgroup patterns

An independent NumPy matrix calculation reconstructs the primary coefficients, HC3 covariance and t-based intervals. Both design matrices have full column rank. No outcome values were changed or trimmed to improve the original estimates.

| Model | N | Independent HC3 SE | Status |
|---|---|---|---|
| primary_time | 346 | 1.140280 | passed |
| primary_quality | 352 | 0.118617 | passed |

## Quality effects at baseline-score quartiles

| Baseline quantile | Writing score | Conditional effect | Pointwise 95% interval |
|---|---|---|---|
| 25% | 3.00 | 0.941 | 0.636 to 1.247 |
| 50% | 4.00 | 0.693 | 0.462 to 0.923 |
| 75% | 5.00 | 0.444 | 0.158 to 0.731 |

These are linear contrasts from the continuous interaction model, not separately randomized subgroup experiments or simultaneous confidence bands. A single baseline task is noisy, and the bounded rating scale can create ceiling effects. The pattern does not establish general employee ability or a workforce-allocation rule.

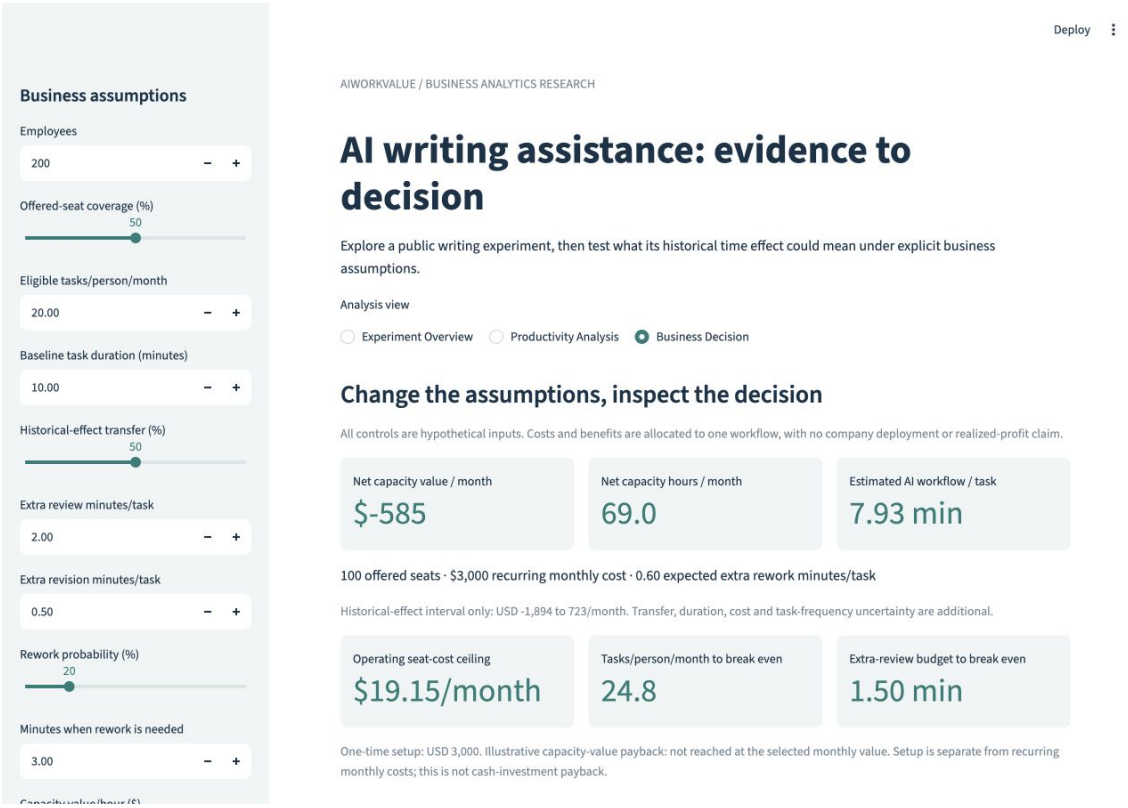
## Three levels of interpretation

Statistical evidence concerns the retained historical assignment contrast. Practical productivity concerns minutes and writing points on particular tasks. Commercial value additionally depends on usable task frequency, duration, checking/rework and cost. None establishes realized company profit.

The primary comparison does not select actual AI users or control for their post-assignment usage. Missing outcomes remain missing, and time outliers are flagged rather than automatically trimmed. Attrition and source exclusions still limit a full-sample causal interpretation.

# Explore the evidence and decision interactively

Streamlit provides Experiment Overview, Productivity Analysis and Business Decision views. The app uses committed aggregate outputs only. Occupation filters affect descriptive comparisons; the full-sample primary coefficients are not silently refitted.




Change task duration, frequency, review, revision, rework probability/time and costs. The baseline cap prevents savings exceeding task duration; the scenario can remain negative. A frequency/review heatmap, interval propagation and break-even metrics update with the controls.

The screenshot uses a 10-minute task, 2 review minutes, 0.5 revision minutes and 0.6 expected rework minutes. Its value differs from the reference case on page 8 because these workflow assumptions differ, not because the historical estimate changed.

Run make dashboard from the project root. GitHub contains the runnable app and actual preview captures; it does not host this Python server. No public hosted deployment or client use is claimed. Automated app tests exercise all three views and check that duration/rework changes update the calculation.